38) What is the struggle for survival?

Darwin noted on the struggle for survival
Struggle for existence inevitably increase variety.
Every being must suffer destruction during some period of its life. Otherwise, on the principal of geometrical increase, no land could support the product

39) What is the principle of natural selection?

population grow exponentially
variation occurs within species (türler arası varyasyon gerçekleşir)
Traits can be innerited. (özellikler atalardan gelir)

40) Explain Mendels work and comment on his due allegence

Mendel work about genetic
He grow and studied 29.000 pea plants, he identified pea plants the 7 pairs of charactenstic for study (form, color, length)

41) Why do you think Morgan and Drosophila in his studies.

Fruit flies have a short life cycle that lasts on average of just 30 days. This flies produce huge number of offspring. Consideration is cost.

42) What are the landmark achievements that led the development of molecular Biology?

The existence of DNA
The elucidation of the structure of DNA
The understanding how DNA controls the structure of the proteins made by cells

43) What is the basis structure of amino acid?

All aminoacids have to the same basic structure. An acid group at are and amine group at the other

45) Briefly explain Protein Structures.

Primary - Exact sequence of amino acids before folding
secondary - Simple folding create simple structure
Tertiary - Folding results in complex 3D structure
Quaternary - multiple 3D subunits organized into a bigger structure.

44) Explain the relationship between biochemistry molecular biology and genetics.

Biochemistry focused on the protein part of life function. It studied the component
Genetics focuses on the gene part. Usually mutants are used.
Molecular Biology integrates these two, as can be quite well ascertained from the "Central Dogma"

SORULAR

29) What are the common elements of organic matters?

Carbon (C), and Hydrogen (H), and Oxygen (O).

30) Different chemical compounds but same composition of element. How do you explain?

When the elements are joined, the atoms lose their individual properties and have different properties from the elements they are composed of. A chemical formula is used a quick way to show the composition of compounds. Letters, numbers and symbols are used to represent elements and the number of elements in each compound.

32) What is the carbon skeleton?

Carbon could have 4 bonds. It forms a chain by connecting to itself or to the other atoms to make organic molekules. Ethane, 2methyl1buttanol.

33) How did the discover of Robert Hooke help the devolepment of biology?

Cells are discovered by Robert Hooke and he investigate cells in 17th century.

34) Name few advances in the microscopy technology?

The electrons microscope advanced biological research.
Lens acting as prism distorts light
20 times as sensitive as the light microscope. (electron microscope)

35) Briefly explain cell theory with respect to plant and animal cell?

All living organisms are composed of cells. They may be unicellular or multicellular. The cell is the basis unit of life. Cells arise from pre-existing cells.

36) What is miasma theory?

The miasma theory states that disease and sicknesses come from breathing in a miasma. It would effect individuals within the locate that gave rise to such vapors.

37) Explain the work of Louis Pasteur and its significance?

Louis Pasteur's pasteurasation experiment illustrate the fact that the spoilage of liquid was caused by particle rather than the air itself. These experiments were important pieces of evidence supporting the idea of Germ Theory of Disease.

BIOLOGY

Scholars of Natural Philosopy concentrated on exploration, discovery and taxonomy. Key to the emergence of modern biology in the 19th century was the development and refinement of light microscope.

Light microscope = This discovery is largely attributed to Robert Hooke and began the scientific study of cells, also known as cell biology.

Nehemiah Grew published several volumes on the Anatomy of Plants exploring various part of plants with the microscope.

Anton van Leeuwenbeck : Lens acting a prism that distort light improved lenses done by Anton Van Leeuwenbeck and later the invention of compound lenses magnified object 500 times. He improved the lenses.

Later the electron microscope marvelously advanced biological research. 20 times as sensitive as the light microscope.

The idea of cells as a fundamental unit of life was eventually formulated in 1839. (Hücrenin bir yaşam birimi olduğu fikri nihayetinde 1839'da formüle edildi)

In the 60's Scheiden believe biology must deal with the actual structure of plants and animals examined.

Schleiden came to conclusion that; the plant cell was the key to understanding botany, and that the nucleus was the universal elementary organ of vegetables.

He concluded that all aspect of plant physiology were fundamentally manifestation of the vital activity of cell.

Schwann studied animals however animal cells were quite transperent therefore it was harder to study.

Like Schlieden, Schwann found the nucleus is the key to understanding the cellular structure of animals

* There was a certain unity in plant and animal cells
* Cells are basis of animal tissue, no matter how specialized.
* All living organisms are composed of one or more cells.
* The cell is the basis unit of structure and organization in organism.

Rudolf Virchow studied medicine and focused on the cellular propagation and concluded that in normal growth, cells propagate from the division of parent cells

Disease cells also grow or multiply from pre-existing cells.

Extensive studied of cancer, convinced virchow that; cancer cells differ from normal cells primarily in behavior rather than in structer. (hücrelerin bozulduğu yapısal değil)

By 1870, scientist identify the first step in cell division in nucleus (mitoz). By the end of the century, using new stoining techiques, chromosomes were discovered and their key function as agent of reproduction were identified.

Disease transmission held that diaseases such as cholera, clomydia or the best black dead were caused by a miasma a noxious form of "Bad air." Such infection was not passed between individuals but would affect individuals within the locate that gave rise to such vapors.

Louis pasteur, John Snow and Robert Koch work of germ and translational period began. (discovery of bacteria)

Louis pasteur's Germ Theory: spoilage of the liquid was caused by particles rather than the air itself, This experiment were important pieces of evidence supporting the idea of germ theory of disease.

Struggle for existence inevitable increase variety. Every begin must suffer destruction during some period of its life. otherwise, on the principal of geometrical increase, no land could support the product.

Organic Chemistry (1800 - 1900)

According to vitalist theory, organic compounds had to be found in nature, they could not be made in the laboratory.

Berzelius → Organic compounds had to be in extracted from nature and then studied in the laboratory. * Berzelius → Vitalist theory

Lavoisier made the first systematic studied of organic composition in 1786. He defined organic chemicals as combinations of oxygen with radicals that contained carbon and hydrogen. He determined carbon content by combustion.

Berzelius analyzed a sufficient number of organik compounds to prove that they obeyed the same laws as inorganic compounds.

combustion= yanma
further= daha ileri
extracted= çıkartılmış
sufficient= yeterli
hence= dolayısıyla
acquired a name= adını almak
strikingly= göze çarpıcı şekilde
percentage= yüzde
determined= belirlemek
arrangement= düzen
set out= yola çıkmak, yapmak

" Aynı bileşime sahip farklı kimyasal bileşiklerin olduğunu ve bunun da farklı yapılara sahip olmak için atomların farklı düzenlemelerine sahip olmaları gerektiğini ima etti. "

Gay Lussac and Thenard found that the ratio of hydrogen to oxygen in sugars, starches, woods and gums was the same as in water, namely two hydrogen atoms for each oxygen atoms. However these compounds did not contain water units. The other main component of these compounds was carbon. Hence, they acquired the name carbonhydrate.

There were different chemical compounds with the same composition which, in turn, implies they must have different arrangements of the atoms to have different structures.

Frearich Wönler (1828) was a young german chemist who set out to make organic compounds in the laboratory. He wished to disprove the Vitalist theory.

In 1837 Aguste Laurent developed the nucleus theory in which there was a three dimensional structure in the form of radical cubes with atoms added to or removed from (of ra) the faces to give different properties.

Valence= değerlik
gain= elde etmek
insight= sezgi

Carbon could have four bonds (Valence)

Friedrich Kekülle (1858) dreamed of carbons forming a chain by connecting to 'useff or to other atoms to make organic molecules.

Benzene gives rise to an important class of compounds that are called aromatic molecules. Due to distinctive smells they are called aromatic

Pasteur observed that certain compounds rotate polarized light. Polarized light was either rotated to the right or to the left due to the arrangement of atoms in the crystal. If coumpound don't rotate polarized light then they were equal mixture of the two forms.

nutshell = gazette

Following dalton, the empirical and theoretical thinking brought logical models which would be the mainstay of chemical understanding.

Wöhler, in a purposeful search to synthesize a natural product succeeded and the vitalist theory that separated organic from inorganic chemistry is explained.

Theories of bonding, which would both explain and predict chemical compounds and properties, were sought.

Because of the foundation of the Scientific Revolution and the Englightenment, important discoveries were made in all areas of science.

During the industrial age, not only was science being used to make material progress but scientific discovery was happening at an unprecedented rate.

The chemical industry, the 19th century was to know hundreds of thousand of new compounds synthesized in the laboratory by organic chemists.

Genetics

mendel (1822-1884)

Genetics was first studied scientifically by Gregor mendel in the 1860's
mendel grew and studied 29,000 pea plants from 1856 to 1863. 7 pairs of characteristic for study
His was a pains-taking researcher that paid close attention to detail
paintaking - ozenu
Mendel discovered that when he crossed one pure pea plants with the other, the hybrid tented to look exactly like one of the parent plants
what mendel observed was that; some characteristics were transmitted virtually intact with little change, while others seemed to disappear, also completely
Mendel was probably the first biologist to work out a detailed application of mathematics to biology
of course, he had no idea of the mechanism by which the pea traits were transmitted.
model organisms are species that are studied to understand the biology of other organism
after humans. Fruit flies share about 1/70 of the genes that cause disease with humans
so scientist can learn about human genetic by studing fruit fly genetics.
They grow and multiply fast (short life cycle) (morgan's group)
morgan's research firmly established mendelian principles with biologists.
Experimental proof of mendelian inferitance would have given importance support to evolutionary theory.
The main focus of genetics has been seek on understanding of the biochemical basis of inheritance.
This fiels is now knows as molecular biology
miller and urey clearly demostrated that the components of proteins could easily be produced from inorganic compounds under natural conditions.

molecular Biology:

Four landmark achievement have occured in molecular biology. These were.

- The existence of DNA
- The elucidation of the structure of DNA
- The understanding how DNA controls the structure of proteins made by cells
- The discovery of the mechanism by which the DNA molecule replicates itself during cell division.

elucidation = izon

Griffith's Experiment:

The first experiment suggesting that bacteria are capable of transferring genetic information through a process known as transformation

X-Ray Crystallography

Bragg's studied the behavior of X rays when shined on crystals. Atoms in crystals cause beam of incident X rays to diffract into many specific directions

diffract into = kırılarak dağılmak

through = vasıtasıyla

Bragg determined a mathematical relationship between the wavelength of the X rays and the distance between the atoms in a crystal through reflection

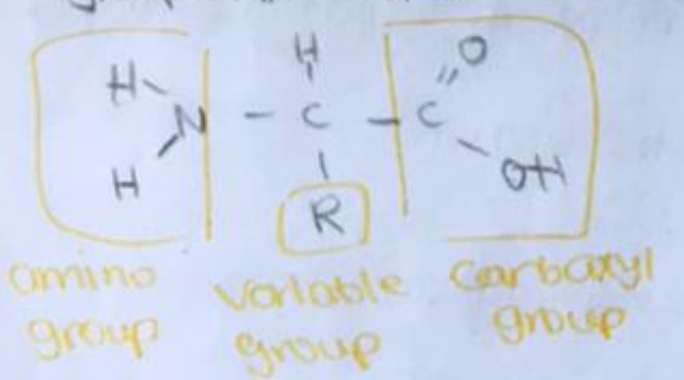
(X ışınlarının dalga boyu ile bir kristalin içindeki atomlar arasındaki uzaklıktan yansıma yoluyla matematiksel bir ilişki belirledi)

Amino Acids:

There are 20 some amino acids found in nature and they all have both an amine end (-NH2) and an acid end (-CO2H)

The resulting molecule is still an amino acid having the characteristic amine and acid ends

All amino acids have the same basic structure, and acid group at one end and an amine group at the other



Proteins:

Proteins are the building blocks of all living cells. (Proteinler tüm canlı hücrelerin yapı taşıdır)

They are large molecules made up of a series of smaller molecules called amino acids.

Aminoacids are linked together to form long chains. It is the order of the aminoacids that determine the particular protein and it's functionally. Typical proteins of importance in the biochemistry of life are made up of hundreds or thousand of amino acids.

After the chains of amino acid are constructed, they can fold into more complex structures.

Genetic code carried by four different bases in DNA; adenin, cytosine, guanine and thymine

It is the order of sets of three of these bases that allows DNA to select individual amino acids in the proper order when assembling proteins

Central Dogma:

Biochemistry focuses on the protein part of life function. It studies the components independent of the organism

Genetic focuses on the gene part. Usually mutants are used.

Molecular Biology integrates those two, as can be quite well as certained form the "Central Dogma)